

My word

Lost knowledge

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We are in the midst of an enormous explosion of new technologies that are making experiments in biology feasible that were unimaginable only a few years ago. These new technologies include significant advances in microscopy, molecular tools, multielectrode recording, methods for monitoring the movements of behaving animals, and of course, the analysis of massive data sets using new and improved statistical methods. But the precipitous pace of our rush forward has a dark side, and is accompanied by a loss of some deep scholarship and expertise that need time to mature in a student.

Many of us senior scientists were trained in traditional disciplines, whether they were physics, chemistry, math, physiology, or biochemistry. Consequently, there was a commitment to a common language within disciplines that could be assumed, at the price of creating knowledge silos. Today, we are witnessing a shift of the pendulum to the other extreme, as ‘interdisciplinarity’ — arguably a common language — is pushed down into undergraduate education so that we are encouraging science students at their earliest stages of education to be conversant across the myriad fields that are now important for many research projects. But at what cost? While our governmental funding agencies are celebrating the value of cross-disciplinary training, do we risk the loss of disciplinary expertise?

Today, for example, there are relatively few trainees who are knowledgeable in theory and practice of membrane biophysics: can you imagine physiologists building their amplifiers today, not to mention designing such? Despite the enormous efforts in determining connectomes — the precise way that neurons are wired together in a network — there are relatively few among us who are training expert electron microscopists. The number

of comparative physiologists and anatomists has decreased, at a time when climate change may require their expertise. There are few left among our scientists who have the expertise in systematics required to reliably identify various species of plants and animals. Machine learning and AI, data sciences, and other high-level approaches have replaced knowledge of the diversity of species and their idiosyncrasies in the hearts and minds of many students.

Some lost knowledge is inconsequential. Probably it does not matter in today’s world just how dial telephones encoded the number called, as simple and elegant as it was. Maybe no one needs to ever again use a slide rule (Figure 1), or even know what it is, although slide rules are a triumph of human ingenuity, and truly beautiful. But surely, there is knowledge that we cannot allow ourselves to lose. For instance, we are sad that so few young people wish to study the theoretical and practical fundamentals of ion channel biophysics and synaptic transmission. We understand the allure of trying to attack the molecular and genetic bases of cognitive processing (if an explanation at that level even exists or makes sense).

But we remain convinced that to judge the relevance of molecular explanations to behavior we will always need people working to understand how molecular biophysics turns into neuronal dynamics. We are not training those people in large numbers today.

We are concerned that the arcane knowledge and subtleties that govern neuronal excitability will be lost in the same way that knowledge of slide rules and dial phones were lost, but that in this instance the lost knowledge will be germane to ‘getting it right’. Likewise, we are concerned that expert knowledge of systematics, comparative anatomy and physiology will be needed again as we confront our planet’s failures and successes in adapting to climate change. But will there be too few who will know the relevant biology, or even will know what they don’t know? Yes, the old books and journals will still exist, but the scientists of the past could never have envisioned the changes of the global environment we are now facing. And one wonders how much the neurologists of 40 years ago knew from physical exams that is not replaced by MRI and CAT scans? As the physicians of the present type into their computers instead of



Figure 1. Slide rule, hourglass and Mollusca fossils (photo by S.M.).

watching their patients, it is hard to know what is lost.

Undergraduates today naturally assume that they will be able to find the answers to anything and everything on their phones; this is the world they grew up in. Therefore they think we are unreasonable if we expect them to remember much of the information that we were required to learn as beginning students. The essential flaw with this strategy is that you cannot reason with understanding or knowledge that you do not know exists, even if your phone knows it. For the phone's remarkable access to the world's knowledge to be helpful, one needs to know what to ask. And it is our existing knowledge that frames the questions that we address to our computers and phones.

Probably every generation has faced the loss of what was considered knowledge when it was supplanted by the creation of new knowledge. Should we mourn all lost knowledge, even when it appears irrelevant? Or, perhaps at the end of the day the more important question is whether the lost knowledge is pertinent to the present and future?

The speed of the development and deployment of Covid diagnostics and therapeutics, despite its unevenness across nationalities, give us reasons to be optimistic. It is a remarkable example of the effectiveness of the enormous explosion of new technologies, even if there are dark sides to these technologies. Perhaps, we should just be grateful that computers and phones are so smart, and that communication across continents is so smooth, and we should simply trust that the next generation will exploit these abilities to quickly work together, and rediscover lost knowledge if they need it. The allure of searching for buried treasure gives us hope that the lost knowledge of our past science will speak to those searching for enlightenment, much as treasure hunters have always been driven to look for other kinds of buried gold.

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Book review

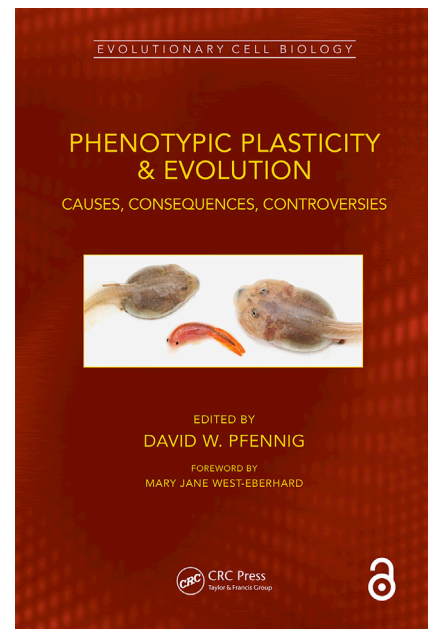
Phenotypic plasticity in a gene-centric world

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Phenotypic Plasticity and Evolution: Causes, Consequences, Controversies
David W. Pfennig, ed.
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The great diversity of forms and functions in nature has fascinated generations of scientists and naturalists. With his seminal book *On the Origin of Species* (1859), Darwin postulated that populations change over time and evolve via natural selection on heritable phenotypic variability. Still, the origin of selectable variation remained a great puzzle to him until the end of his life, and this is expressed in his statement, “there is hardly any question in biology of more importance than this of nature and causes of variability” (quoted in chapter 2 by Costa and from¹). Since then, the field of evolutionary biology has advanced greatly on resolving this issue, especially after the discovery of genes and how they tend to combine with environmental effects to create phenotypic variation. Nevertheless, even though it is generally recognized that the environment provides a major contribution to the phenotype (i.e. most traits have modest but non-zero heritability)², how such environmental effects are relevant for evolutionary processes remains controversial.

Phenotypic plasticity, “the ability of a given genotype to express different phenotypes in different environmental circumstances” (Sultan, p. 3), has long been acknowledged by evolutionary biologists and dates back to debates before the modern synthesis. The field of quantitative genetics naturally incorporates plasticity into its thinking, taking the genotype–environment interaction into account as part of phenotype formation, and as a factor that itself can evolve. That is, quantitative genetics considers that a single genotype can respond



differently to a range of environmental pressures (i.e. the reaction norm), thus creating a range of phenotypes. However, quantitative genetics has been overshadowed, at least in some contexts, by population genetics in publications and in general attention. For example, most formal theories preceding the modern synthesis were population genetics models³, and empirical interest in genes has blossomed following the advent of next-generation sequencing⁴. Population genetics in turn has not frequently incorporated phenotypic plasticity into its models, having traditionally focused on alleles and genes. This ‘gene-centric perspective’ has been put into question by several researchers, and this has led to heated debates concerning the need to better include plasticity, developmental processes, and epigenetics into evolutionary thinking⁵. In many situations, such inflamed positions have left unclear how to incorporate plasticity into a gene-centered view of evolution.

In this context, the book *Phenotypic Plasticity and Evolution: Causes, Consequences, Controversies* by David Pfennig and colleagues helps to clarify that plasticity is not so incompatible with genetic evolution. The book introduces relevant concepts from quantitative genetics and other fields as well as provides